

SatQuery AI — What This Project Actually Is

Written for someone seeing this project for the very first time.

The one-sentence idea

SatQuery AI lets you ask a satellite picture a question in plain English — "what changed here?", "is there water in this area?", "what kind of land is this?" — and get back a real answer, along with a step-by-step explanation of exactly how the system arrived at it.

The problem this solves

Organisations like ISRO produce enormous amounts of satellite imagery every day — optical photos (like a camera from space) and radar scans (which can see through clouds and at night, but look nothing like a normal photo). Reading these images usefully requires trained specialists: knowing which of dozens of colour bands to look at, how to fix a radar image so it isn't just a black square, how to compare two photos of the same place taken months apart, and so on.

Most people — a policymaker, a disaster-response coordinator, a farmer, a student — can't do any of that. They just have a question. SatQuery AI is the translator in between: you ask in plain words, it does the technical work, and it hands back an answer a non-expert can actually use.

Where this idea comes from

This is being built for Smart India Hackathon 2026, problem statement **SIH26167**, posed by ISRO: build an "agentic vision-language assistant" for satellite imagery — software that doesn't just pattern-match pixels, but reasons through a request the way a human analyst would: figure out what's being asked, pick the right tool for the job, run it, and explain the result.

The 5 things it has to be able to do

ISRO's brief lists five required abilities. In plain terms:

- **1. Actually understand satellite images, not just normal photos.** A satellite image isn't a JPEG — it can have 12+ colour channels invisible to the human eye, or be radar data that looks like static until it's processed correctly.
- **2. Answer questions about one image, and describe what's in it.** "What is this?", "describe this scene", "is there a road here?" — the basic Q&A/captioning ability.
- **3. Compare two images of the same place over time.** Feed it a "before" and "after" picture and it tells you what changed — useful for tracking floods, construction, deforestation, disaster damage.
- **4. Combine two different sensor types for a better answer.** Optical images show colour and detail but are useless through cloud cover; radar sees through clouds but looks alien. Using both together ("fusion") should genuinely beat using just one.
- **5. Show its work.** Not a black box — every answer must come with a visible, checkable trail of exactly what the system decided and why, so a human can trust (or catch a mistake in) the answer.

Why "showing its work" is the actual centrepiece

This is arguably the most important idea in the whole project, and the easiest one to miss. Most AI demos show you a slick answer and ask you to trust it. SatQuery AI is built to do the opposite: every single query prints out its full reasoning trail — what image it read, whether the image passed quality checks, what kind of question it thinks this is, which specific tool it picked (and, importantly, which tools it considered and rejected, and why), what that tool actually found, and finally a confidence score broken into four separate named reasons rather than one vague percentage.

That matters because this is meant for a government/space-agency context — a system that quietly guesses and hopes is not something anyone can rely on for, say, disaster response or land monitoring. "Trustworthy because it's checkable" is the actual pitch, not "trustworthy because it's usually right."

What actually happens when you use it

Picture the finished product as a simple webpage:

- You drag in one or two satellite images.
- You type a question in plain English.
- You hit "Ask."
- Within half a second, lines start appearing one at a time on screen — each one a step the system just finished ("read the image", "checked it's valid", "figured out this is a change-detection question", "picked this specific AI tool because...", "ran it", "here's what it found"), building up in real time rather than appearing all at once at the end.
- Alongside the trace, you see the images themselves rendered properly (including turning a raw radar file from an unreadable black square into something you can actually look at), and — for change questions — a highlighted overlay showing exactly what changed.
- Finally, a plain-English answer appears, with a confidence score you can click into to see exactly why the system is (or isn't) sure.

How it works under the hood (without the jargon)

Two deliberate design choices shape everything else:

- **The AI decides "what kind of question is this", but plain code decides "which tool handles it."** If tool selection were left to an AI's judgement, the same question could get answered two different ways on two different runs — bad for something that needs to be checkable. So a simple, predictable rulebook (a registry of tools, each labelled with what it can and can't do) makes that decision instead, and that decision is always the same given the same input.
- **Comparing two images is broken into separate, visible steps** rather than one AI model doing everything invisibly: first a detector finds what pixels changed, then plain arithmetic turns that into numbers ("2.4% of the scene changed"), then a language model turns those numbers into a sentence a human reads. Every one of those steps shows up separately in the trace.

Being honest about what's real today vs. what's still a stand-in

This is a hackathon prototype, not a finished product, and the plan has always been explicit about that rather than hiding it:

- The overall pipeline, the website, the reasoning trace, the tool-selection system, and the SAR image-fixing are all real and fully working today.

- One real trained AI model exists so far — it looks at a satellite patch and identifies what kind of land it is (forest, farmland, water, city, etc.), and has already proven that combining optical + radar images gives a better answer than either alone.
- A few other pieces (answering free-form questions in full sentences, and the change-detection step specifically) currently use an honest placeholder that's clearly labelled as such in the trace, standing in for a trained model that hasn't been built yet — rather than faking a confident-sounding answer with no real model behind it.

What "done" looks like for the hackathon

A working demo where a judge can hand over two real satellite images they've never seen before, ask a genuine question, and watch the system think out loud in real time before giving an answer it can actually defend — with the honest caveat, stated up front rather than discovered later, of exactly which parts are fully trained and which are clearly-labelled placeholders for future work.